Using DeMorgan's law we get
$\neg female(sita) \vee \neg sister(sita, geetha) \vee \neg parent(geetha, mohan)$
$\vee aunt(sita, mohan)$ — IR_1 (Instantiation of rule 1)
The conclusion we want to derive is aunt(sita, mohan).
Negating the conclusion we get $\neg aunt(sita, mohan)$ — NC (negation of conclusion)

From F_1, F_2, F_3, IR_1 and NC, it can be seen that using resolution we can derive the empty clause. Hence the conclusion aunt(sita, mohan) is correct.

- The goal is aunt(sita, mohan) and Prolog tries to find rules and facts and using proper instantiation tries to see whether the goal is satisfied. This is called backward chaining.

Thus we see that the resolution principle is used in the logic programming language Prolog. It is also used in automatic theorem proving where a program or software package is used to prove theorems, given the axioms.

1.6.7 Fallacies

- Studying common fallacies help us distinguish between correct and incorrect reasoning.
- The proposition $\boxed{((p \rightarrow q) \wedge q) \rightarrow p}$ is not a tautology, because it is false when p is false and q is true. However, there are many incorrect arguments that treat this as a tautology. In other words they treat the argument with premises $p \rightarrow q$ and q and conclusion p as a valid argument form, which it is not. This type of incorrect reasoning is called the $\boxed{\text{fallacy of affirming the conclusion}}$.